

Solution Requirements

Date	6 July 2026
Team ID	SWTID-2026-8977
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely access EduGenie. Optional login using email/Google account for saving learning history and personalized recommendations.	High
2	Authorization levels	Role-based access: Student/Learner can ask questions, generate quizzes, summaries, and learning paths. Administrator can manage system settings, monitor usage, and maintain the platform.	Medium

3	External interfaces	Integrates with Google Gemini API for AI-generated educational responses. FastAPI communicates with the frontend using REST APIs.	High
4	Transactions processing	User requests (questions, quiz generation, summaries, learning paths) are processed by the FastAPI backend, sent to the Gemini API, and the AI-generated responses are returned to the user.	High
5	Reporting	Displays generated answers, quizzes, summaries, personalized learning paths, and optionally stores user learning history and progress reports.	Medium
6	Business rules, etc.	The system validates user input, generates only educational content, provides concise and accurate responses, and supports multiple learning features without storing sensitive personal data.	High
7	Compliance to laws or regulations	The system follows data privacy best practices and complies with applicable privacy regulations (e.g., GDPR principles) while protecting user information and API credentials.	Medium
8	<i>Other</i>	Provides intelligent concept explanations, educational text summarization, AI-powered quiz generation, and personalized learning	High

		recommendations through an interactive user interface.	
--	--	--	--

Step 2: Non-Functional Requirements (NFR)

1	Performance & Speed	The system should process user requests quickly and display AI-generated responses with minimal delay.	Average response time < 3 seconds (depending on API response)
2	Scalability	The application should support increasing numbers of users and learning requests without significant performance degradation.	Supports 500+ concurrent users
3	Security & Data Privacy	Secure API communication using HTTPS, protect API keys, validate user input, and avoid storing sensitive personal information.	HTTPS enabled, secure API key management, input validation
4	Reliability & Availability	The system should remain available with minimal downtime and handle API failures gracefully.	99.5% uptime with proper error handling
5	Usability & Accessibility	Provide an intuitive, responsive, and easy-to-use interface accessible across desktops, laptops, tablets, and mobile devices.	Responsive UI with user-friendly navigation

6	<i>Other</i>	The system should have a modular architecture, be easy to maintain, and support deployment on Windows, macOS, and Linux environments.	Modular FastAPI architecture with cross-platform compatibility
---	--------------	---	--